Azure AI

Azure AI Language

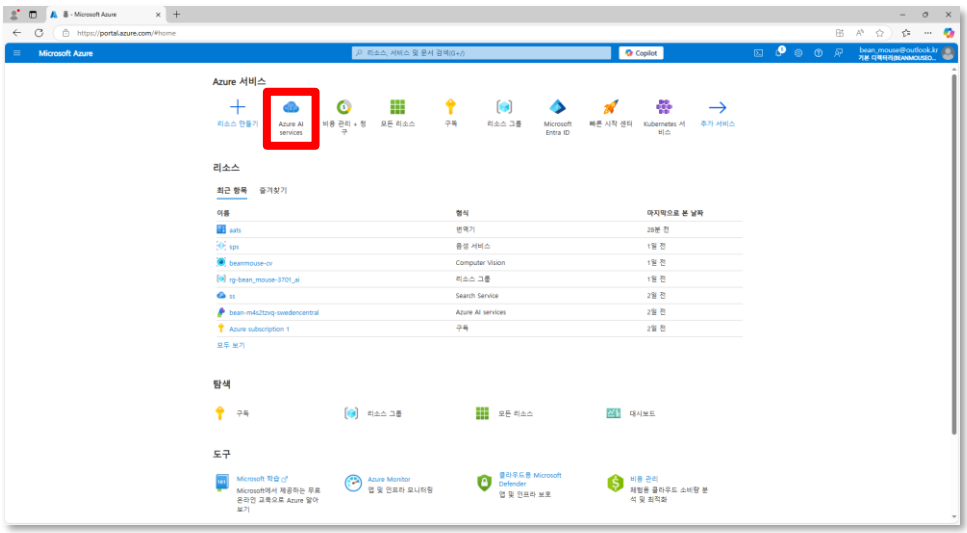
권기웅

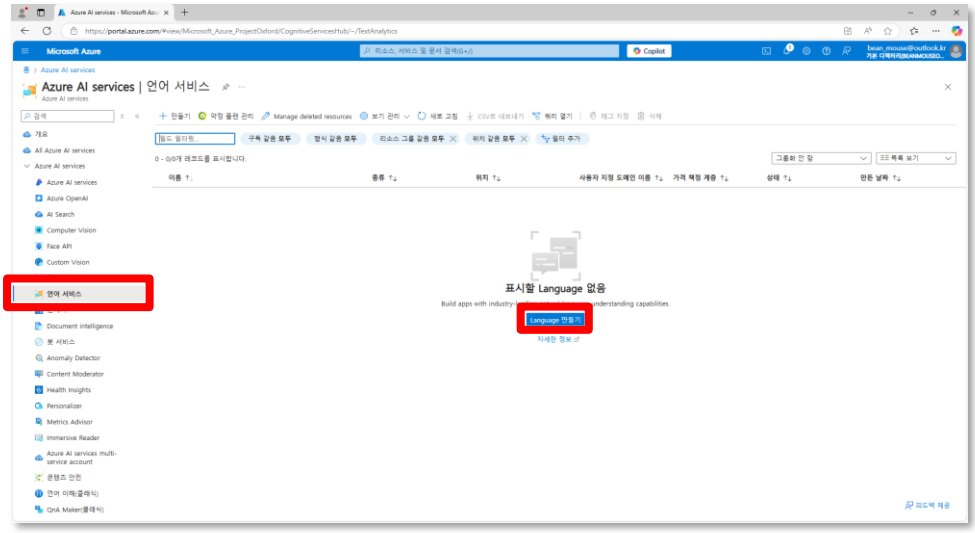
1. 환경설정

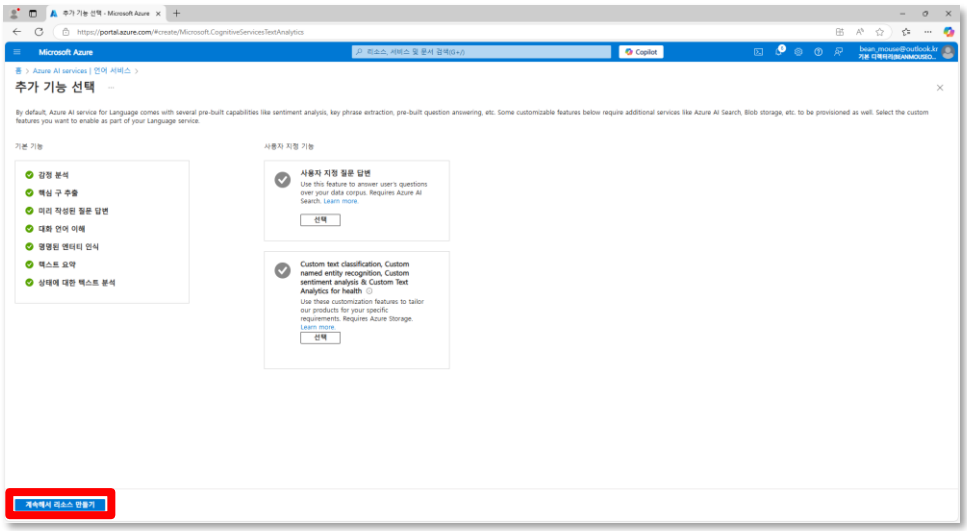
1.1 Azure AI Language 리소스

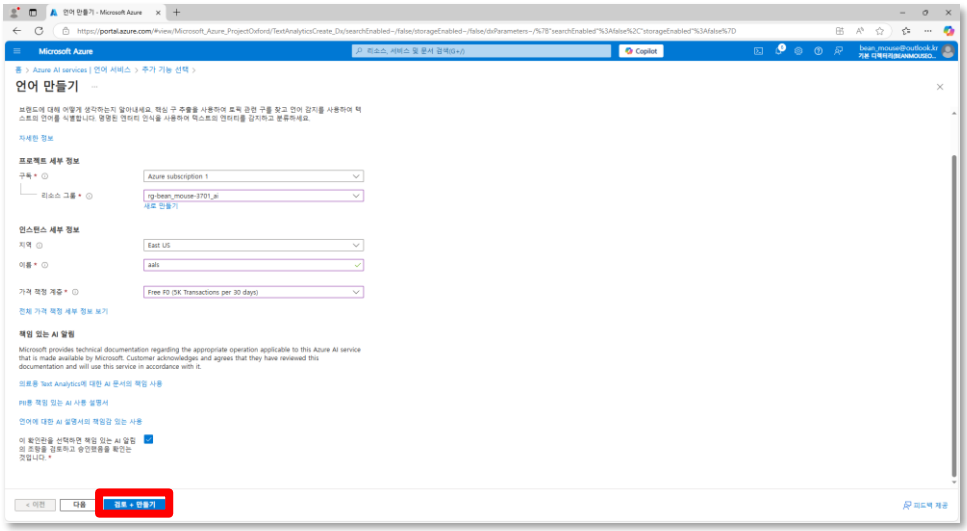
1.1.1 리소스 만들고

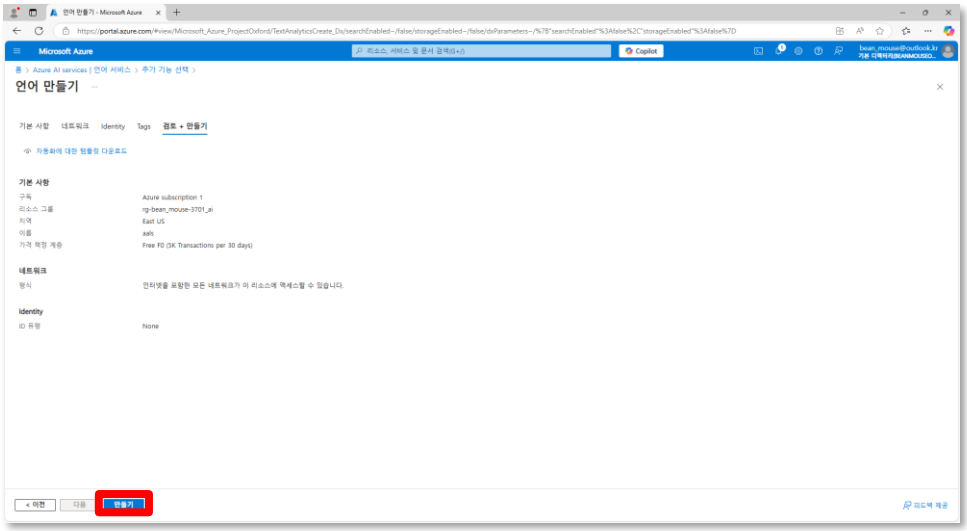
https://portal.azure.com 에서



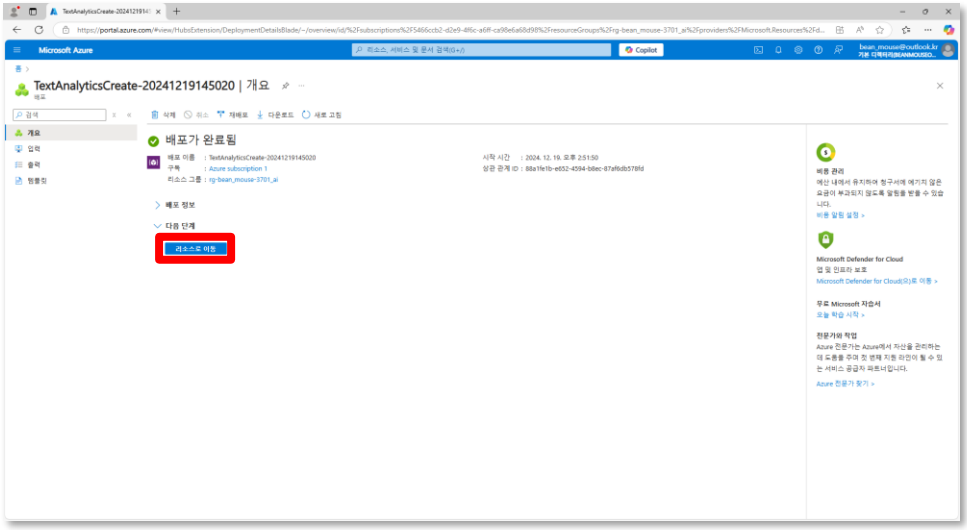


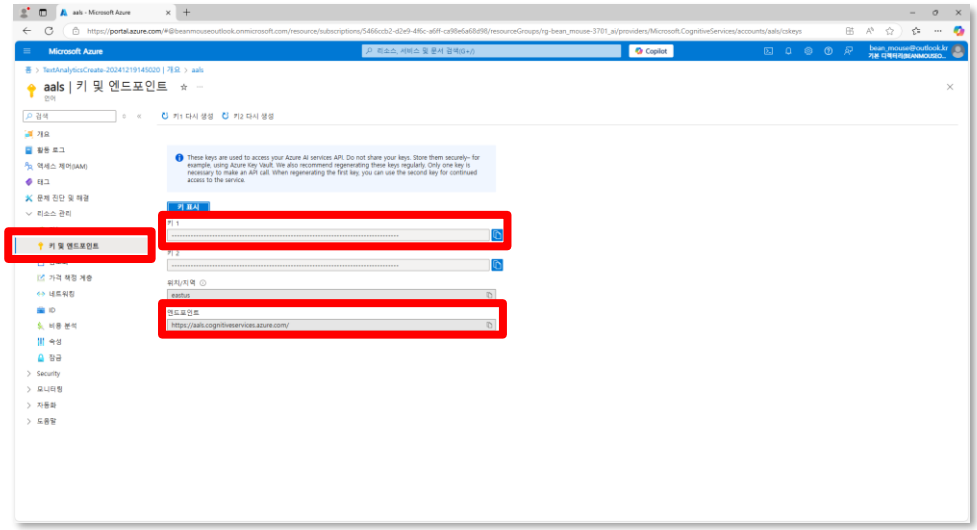

추가 기능 선택 - Microsoft Azure
Microsoft Azure
홈 > Azure AI services | 언어 서비스 >
추가 기능 선택
By default, Azure AI service for Language comes with several pre-built capabilities like sentiment analysis, key phrase extraction, pre-built question answering, etc. Some customizable features below require additional services like Azure AI Search, Blob storage, etc. to be provisioned as well. Select the custom features you want to enable as part of your Language service.
기본 기능
감정 분석
핵심 구 추출
미리 작성된 질문 답변
대화 언어 이해
명명된 엔티티 인식
텍스트 요약
상태에 대한 텍스트 분석
사용자 지정 기능
사용자 지정 질문 답변
Use this feature to answer user's questions over your data corpus. Requires Azure AI Search. Learn more.
선택
Custom text classification, Custom named entity recognition, Custom sentiment analysis & Custom Text Analytics for health
Use these customization features to tailor our products for your specific requirements. Requires Azure Storage.
Learn more.
선택
계속해서 리소스 만들기


언어 만들기 - Microsoft Azure
Microsoft Azure
홈 > Azure AI services | 언어 서비스 > 추가 기능 선택 >
언어 만들기
브랜드에 대해 어떻게 생각하는지 알아내세요. 핵심 구 추출을 사용하여 트윗 관련 구를 찾고 언어 감지를 사용하여 텍스트의 언어를 식별합니다. 명명된 엔티티 인식을 사용하여 텍스트의 엔티티를 감지하고 분류하세요.
자세한 정보
프로젝트 세부 정보
구독 *
Azure subscription 1
리소스 그룹 *
rg-bean_mouse-3701_ai
새로 만들기
인스턴스 세부 정보
지역
East US
이름 *
aals
가격 책정 계층 *
Free F0 (5K Transactions per 30 days)
전체 가격 책정 세부 정보 보기
책임 있는 AI 알림
Microsoft provides technical documentation regarding the appropriate operation applicable to this Azure AI service that is made available by Microsoft. Customer acknowledges and agrees that they have reviewed this documentation and will use this service in accordance with it.
의료용 Text Analytics에 대한 AI 문서의 책임 사용
PII용 책임 있는 AI 사용 설명서
언어에 대한 AI 설명서의 책임감 있는 사용
이 확인란을 선택하면 책임 있는 AI 알림의 조항을 검토하고 승인했음을 확인는 것입니다. *
< 이전
다음
검토 + 만들기
피드백 제공


언어 만들기 - Microsoft Azure
Microsoft Azure
홈 > Azure AI services | 언어 서비스 > 추가 기능 선택 >
언어 만들기
기본 사항
네트워크
Identity
Tags
검토 + 만들기
자동화에 대한 템플릿 다운로드
기본 사항
구독
Azure subscription 1
리소스 그룹
rg-bean_mouse-3701_ai
지역
East US
이름
aals
가격 책정 계층
Free F0 (5K Transactions per 30 days)
네트워크
형식
인터넷을 포함한 모든 네트워크가 이 리소스에 액세스할 수 있습니다.
Identity
ID 유형
None
< 이전
다음
만들기
피드백 제공

1.1.2 키, 엔드포인트 챙기기

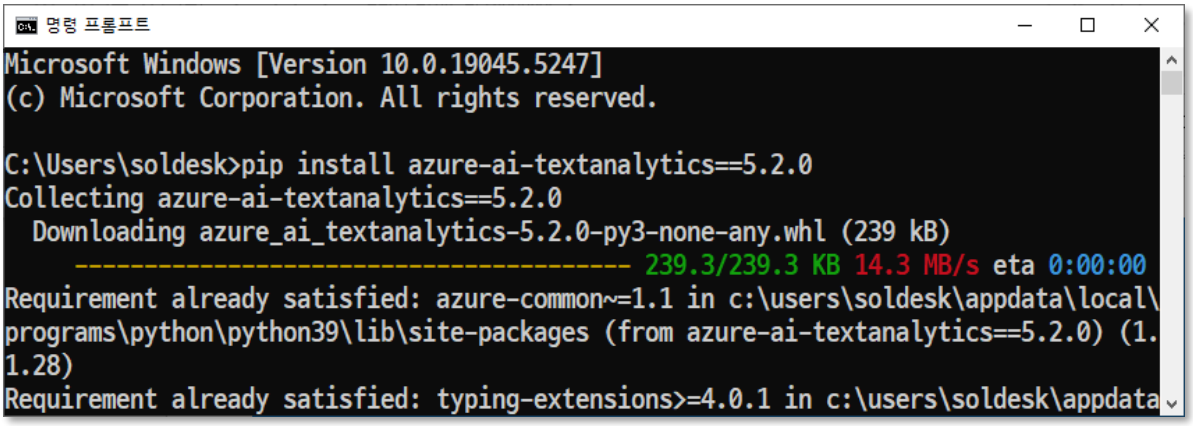




2. Entity 링크(영어만)

2.1 Python 라이브러리 설치

```
pip install azure-ai-textanalytics==5.2.0
```



2.2 Python 프로그램

```python
# -*- coding:utf-8 -*-
from azure.ai.textanalytics import TextAnalyticsClient
from azure.core.credentials import AzureKeyCredential

language_key = "AXvKbyDVuh1HBKFx3TNnAqjCoZ4PDKls9XSAhSUNgwKgAMJrszj9JQQJ99ALACYeBjFXJ3w3AAAaACOGxurz"  # API키
language_endpoint = "https://aals.cognitiveservices.azure.com/"  # 엔드포인트

text_analytics_client = TextAnalyticsClient(
    endpoint=language_endpoint, credential=AzureKeyCredential(language_key)
)

documents = [
    """To Sherlock Holmes she is always _the_ woman. I have seldom heard him
mention her under any other name. In his eyes she eclipses and
predominates the whole of her sex. It was not that he felt any emotion
akin to love for Irene Adler. All emotions, and that one particularly,
were abhorrent to his cold, precise but admirably balanced mind. He
was, I take it, the most perfect reasoning and observing machine that
the world has seen, but as a lover he would have placed himself in a
false position. He never spoke of the softer passions, save with a gibe
and a sneer. They were admirable things for the observer—excellent for
drawing the veil from men's motives and actions. But for the trained
reasoner to admit such intrusions into his own delicate and finely
adjusted temperament was to introduce a distracting factor which might
throw a doubt upon all his mental results. Grit in a sensitive
instrument, or a crack in one of his own high-power lenses, would not
be more disturbing than a strong emotion in a nature such as his. And
yet there was but one woman to him, and that woman was the late Irene
Adler, of dubious and questionable memory.."""
]
result = text_analytics_client.recognize_linked_entities(documents=documents)[0]

for entity in result.entities:
    print(
        "name : %s, id : %s, url : %s(%s)"
        % (entity.name, entity.data_source_entity_id, entity.url, entity.data_source)
    )
print("matches:")
for match in entity.matches:
    print("\ttext: %s, conf : %.2f" % (match.text, match.confidence_score))
```

```
PS D:\Kwon\workspaceMS>  d:; cd 'd:\Kwon\workspaceMS'; & 'c:\Users\soldesk\AppData\Local\Programs\Python\Pyt
24.14.0-win32-x64\bundled\libs\debugpy\adapter/../..\debugpy\launcher' '50576' '--' 'd:\Kwon\workspaceMS\ent
name : Sherlock Holmes, id : Sherlock Holmes, url : https://en.wikipedia.org/wiki/Sherlock_Holmes(Wikipedia)
name : Irene Adler, id : Irene Adler, url : https://en.wikipedia.org/wiki/Irene_Adler(Wikipedia)
matches:
        text: Irene Adler, conf : 0.45
        text: Irene, conf : 0.45
        text: Adler, conf : 0.45
PS D:\Kwon\workspaceMS>
```

3. 언어 감지

3.1 Python 프로그램

```python
# -*- coding:utf-8 -*-
from azure.ai.textanalytics import TextAnalyticsClient
from azure.core.credentials import AzureKeyCredential

language_key = "AXvKbyDVuh1HBKFx3TNnAqjCoZ4PDKls9XSAhSUNgwKgAMJrszj9JQQJ99ALACYeBjFXJ3w3AAAaACOGxurz"  # API키
language_endpoint = "https://aals.cognitiveservices.azure.com/"  # 엔드포인트

text_analytics_client = TextAnalyticsClient(
    endpoint=language_endpoint, credential=AzureKeyCredential(language_key)
)

documents = ["權", "기", "ung"]
for l in text_analytics_client.detect_language(documents=documents):
    print(l.primary_language.name)
```

```
PS D:\Kwon\workspaceMS>  d:; c
ions\ms-python.debugpy-2024.14
Language:  Chinese_Simplified
Language:  Korean
Language:  English
```

4. 핵심 구 추출

4.1 Python 프로그램

```python
# -*- coding:utf-8 -*-
from azure.ai.textanalytics import TextAnalyticsClient
from azure.core.credentials import AzureKeyCredential

language_key = "AXvKbyDVuh1HBKFx3TNnAqjCoZ4PDKls9XSAhSUNgwKgAMJrszj9JQQJ99ALACYeBjFXJ3w3AAAaACOGxurz"  # API키
language_endpoint = "https://aals.cognitiveservices.azure.com/"  # 엔드포인트

text_analytics_client = TextAnalyticsClient(
    endpoint=language_endpoint, credential=AzureKeyCredential(language_key)
)

documents = [
    "손흥민(32·토트넘 홋스퍼)과 해리 케인(31·바이에른 뮌헨)의 재결합 가능성에 독일과 영국의 시선이 쏠리고 있다."
]
response = text_analytics_client.extract_key_phrases(documents=documents)[0]

for phrase in response.key_phrases:
    print(phrase)
```

```
PS D:\Kwon\workspaceMS>  d:; cd '
ions\ms-python.debugpy-2024.14.0-
손
트
리
케인
이
가
성
과
영국의
```

5. NER
(Named Entity Recognition)

5.1 Python 프로그램

```python
# -*- coding:utf-8 -*-
from azure.ai.textanalytics import TextAnalyticsClient
from azure.core.credentials import AzureKeyCredential

language_key = "AXvKbyDVuh1HBKFx3TNnAqjCoZ4PDKls9XSAhSUNgwKgAMJrszj9JQQJ99ALACYeBjFXJ3w3AAAaACOGxurz"  # API키
language_endpoint = "https://aals.cognitiveservices.azure.com/"  # 엔드포인트

text_analytics_client = TextAnalyticsClient(
    endpoint=language_endpoint, credential=AzureKeyCredential(language_key)
)

documents = [
    """
  누상촌 현덕은 원래가 황손
  관우 장비와 의를 맺어 천자 돕기 원했으나
  동서로 뛰어다녀도 근거할 땅 없는데다
  장수 적고 졸개 모자라니 떠돌이신세였다.
  남양 땅 세 번 찾으니 그 정 얼마나 깊은가
  와룡선생은 한눈에 천하의 나뉨을 알아보네
  먼저 형주 뺏고 뒤에 서천 차지하니
  패업과 임금의 길 거기에 있었다.
"""
]
result = text_analytics_client.recognize_entities(documents=documents)[0]

for entity in result.entities:
    print(
        "text : %s, cate : %s, subCate : %s, conf : %.2f, len : %d, offset : %d"
        % (
            entity.text,
            entity.category,
            entity.subcategory,
            entity.confidence_score,
            entity.length,
            entity.length,
        )
    )
```

```
PS D:\Kwon\workspaceMS>  d:; cd 'd:\Kwon\workspaceMS'; & 'c:\Users\soldesk\AppData\Loc
ions\ms-python.debugpy-2024.14.0-win32-x64\bundled\libs\debugpy\adapter/../..\debugpy\
text : 누상촌 현덕, cate : Location, subCate : None, conf : 0.80, len : 6, offset : 6
text : 동서, cate : Location, subCate : None, conf : 0.53, len : 2, offset : 2
text : 자, cate : Product, subCate : None, conf : 0.53, len : 1, offset : 1
text : 남양, cate : Location, subCate : None, conf : 0.90, len : 2, offset : 2
text : 와룡, cate : Person, subCate : None, conf : 0.96, len : 2, offset : 2
text : 형주, cate : Location, subCate : GPE, conf : 0.74, len : 2, offset : 2
text : 서천, cate : Location, subCate : GPE, conf : 0.74, len : 2, offset : 2
text : 업, cate : Skill, subCate : None, conf : 0.67, len : 1, offset : 1
```

6. PII

(Personally Identifying Information)

6.1 Python 프로그램

```python
# -*- coding:utf-8 -*-
from azure.ai.textanalytics import TextAnalyticsClient
from azure.core.credentials import AzureKeyCredential

language_key = "AXvKbyDVuh1HBKFx3TNnAqjCoZ4PDKls9XSAhSUNgwKgAMJrszj9JQQJ99ALACYeBjFXJ3w3AAAaACOGxurz"  # API키
language_endpoint = "https://aals.cognitiveservices.azure.com/"  # 엔드포인트

text_analytics_client = TextAnalyticsClient(
    endpoint=language_endpoint, credential=AzureKeyCredential(language_key)
)

documents = [
    "권기웅, 수석강사/학과장, 010-3154-4435, bean_mouse@soldesk.com, 서울특별시 종로구 종로12길 15(관철동) 종로코아빌딩 [2F,5F,8F,9F,10F]"
]
response = text_analytics_client.recognize_pii_entities(documents)
for info in response:
    print("기밀 정보 :", info.redacted_text)
    for entity in info.entities:
        print(
            "entity : %s, cate : %s, conf : %.2f, offset : %d, len : %d"
            % (
                entity.text,
                entity.category,
                entity.confidence_score,
                entity.offset,
                entity.length,
            )
        )
```

```
PS D:\Kwon\workspaceMS>  d:; cd 'd:\Kwon\workspaceMS'; & 'c:\Users\soldesk\AppData\Local\Programs\Python\Python39\python
24.14.0-win32-x64\bundled\libs\debugpy\adapter/../..\debugpy\launcher' '50045' '--' 'd:\Kwon\workspaceMS\pii.py'
기밀 정보 : ***, ****/***, **********, **********, **********(***) 종로코아빌딩 [2F,5F,8F,9F,10F]
entity : 권기웅, cate : Person, conf : 1.00, offset : 0, len : 3
entity : 수석강사, cate : PersonType, conf : 0.73, offset : 5, len : 4
entity : 학과장, cate : PersonType, conf : 0.81, offset : 10, len : 3
entity : 010-3154-4435, cate : PhoneNumber, conf : 0.80, offset : 15, len : 13
entity : bean_mouse@soldesk.com, cate : Email, conf : 0.80, offset : 30, len : 22
entity : 서울특별시 종로구 종로12길 15, cate : Address, conf : 0.78, offset : 54, len : 18
entity : 관철동, cate : Address, conf : 0.57, offset : 73, len : 3
```

7. 감정 분석

7.1 Python 프로그램

```python
# -*- coding:utf-8 -*-
from azure.ai.textanalytics import TextAnalyticsClient
from azure.core.credentials import AzureKeyCredential

language_key = "AXvKbyDVuh1HBKFx3TNnAqjCoZ4PDKls9XSAhSUNgwKgAMJrszj9JQQJ99ALACYeBjFXJ3w3AAAaACOGxurz"  # API키
language_endpoint = "https://aals.cognitiveservices.azure.com/"  # 엔드포인트

text_analytics_client = TextAnalyticsClient(
    endpoint=language_endpoint, credential=AzureKeyCredential(language_key)
)

documents = [
    "은비대리가 춥다고 내 핫팩을 훔쳤다. 도둑이다. 날강도가 따로 없다. 아주 못됐다."
]

result = text_analytics_client.analyze_sentiment(documents, show_opinion_mining=True)

for document in result:
    print("전체 : %s" % (document.sentiment), end="")
    print(
        "(긍정=%.2f, 중립=%.2f, 부정=%.2f)"
        % (
            document.confidence_scores.positive,
            document.confidence_scores.neutral,
            document.confidence_scores.negative,
        )
    )
    print("문장")
    for sentence in document.sentences:
        print("\t%s: %s" % (sentence.text, sentence.sentiment), end="")
        print(
            "(긍정=%.2f, 중립=%.2f, 부정=%.2f)"
            % (
                sentence.confidence_scores.positive,
                sentence.confidence_scores.neutral,
                sentence.confidence_scores.negative,
            )
        )
```

```
PS D:\Kwon\workspaceMS>  d:; cd 'd:\Kwon\workspaceMS'; & 'c:\Users\soldesk\AppData\Local
ions\ms-python.debugpy-2024.14.0-win32-x64\bundled\libs\debugpy\adapter/../..\debugpy\la
전체 : negative(긍정=0.01, 중립=0.12, 부정=0.87)
문장
        은비대리가 춥다고 내 핫팩을 훔쳤다. : negative(긍정=0.01, 중립=0.23, 부정=0.77)
        도둑이다. : neutral(긍정=0.03, 중립=0.60, 부정=0.37)
        날강도가 따로 없다. : neutral(긍정=0.01, 중립=0.96, 부정=0.03)
        아주 못됐다.: negative(긍정=0.01, 중립=0.02, 부정=0.97)
PS D:\Kwon\workspaceMS>
```

8. 의료

8.1 Python 프로그램

```python
# -*- coding:utf-8 -*-
from azure.ai.textanalytics import TextAnalyticsClient
from azure.core.credentials import AzureKeyCredential

language_key = "AXvKbyDVuh1HBKFx3TNnAqjCoZ4PDKls9XSAhSUNgwKgAMJrszj9JQQJ99ALACYeBjFXJ3w3AAAaACOGxurz"  # API키
language_endpoint = "https://aals.cognitiveservices.azure.com/"  # 엔드포인트

text_analytics_client = TextAnalyticsClient(
    endpoint=language_endpoint, credential=AzureKeyCredential(language_key)
)

documents = [
    "타이레놀 2개씩 아플때마다 먹고, 후시딘 샤워하고나서 조금씩 상처난데 바르고, 구내염 생긴곳에 알보칠 조금 발라서 지져요",
    "감기에는 약이 없으니 라면이나 끓여먹어요",
]

poller = text_analytics_client.begin_analyze_healthcare_entities(documents)
result = poller.result()

for doc in result:
    for entity in doc.entities:
        print(entity.text)
        print("\tnormalized text :", entity.normalized_text)
        print("\tcate :", entity.category)
        print("\tsubCate :", entity.subcategory)
        print("\toffset :", entity.offset)
        print("\tconf :", entity.confidence_score)
    for relation in doc.entity_relations:
        print(relation.relation_type)
        for role in relation.roles:
            print("\t%s : %s" % (role.name, role.entity.text))
```

```
구내염 생긴곳에
        normalized text : None
        cate : SymptomOrSign
        subCate : None
        offset : 44
        conf : 0.82
알보칠
        normalized text : None
        cate : SymptomOrSign
        subCate : None
        offset : 53
        conf : 0.94
조금
        normalized text : None
        cate : ConditionQualifier
        subCate : None
        offset : 57
        conf : 0.85
발라서
        normalized text : None
        cate : SymptomOrSign
        subCate : None
        offset : 60
        conf : 0.86
DosageOfMedication
        Medication : 타이레놀
        Dosage : 2개씩
RouteOfMedication
        Medication : 후시딘
```